Naveed Ahmed

Full Stack AI Engineer | Lahore, Pakistan

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portfolio-naveed.vercel.app

Professional Summary

Full Stack AI Engineer with 4+ years of experience building scalable web applications, enterprise software, and production AI systems. Currently building an AI executive assistant at Leaders Performance (Dubai, remote), with document-grounded answers that cite their sources, bilingual English and Dutch voice, reliability on unstable networks, and access control on every read and write. Hands-on with LLM integration, Retrieval-Augmented Generation (RAG), AI agents with tool calling, structured LLM outputs, real-time streaming, and voice input, on a full-stack foundation. Experienced in leading development teams, acting as the primary technical contact for clients, and taking products from concept to production across aviation, inventory management, digital media, and AI.

Technical Skills

Languages: HTML, CSS, JavaScript, TypeScript, Python
Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, NestJS, Django, Prisma, Zod, Redux Toolkit, Tailwind CSS, Radix UI, Bootstrap
AI Engineering: LLM Integration, Google Gemini, Generative AI, AI Agents, Tool Calling, RAG, Embeddings, Vector Search, Structured Output, SSE Streaming, Speech Recognition, Text-to-Speech, Multilingual Voice, Agent Memory, Human-in-the-Loop, Guardrails, Recommendations
Databases: MongoDB, MySQL, PostgreSQL (pgvector), Supabase
Messaging & Caching: Redis, RabbitMQ, BullMQ
Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Microservices, API Gateways, CI/CD Pipelines
APIs & Integrations: REST and RPC APIs, Swagger/OpenAPI, OAuth, Lark Open Platform
Testing & Quality: Vitest, Testing Library, Playwright, Accessibility (WCAG, axe-core), Database Migrations with Rollback
Tools: Git, GitHub, Postman, JIRA, ClickUp, Claude, Cursor

Experience

Leaders Performance — October 2026 – Present

Full Stack AI Engineer — *Dubai, UAE (Remote)*

- Building an **AI Executive Assistant** for the company's leadership as a full-stack engineer, taking features from technical design to production-ready releases.
- Designing AI agent architecture and LLM-driven workflows with human review and approval built into the flow.
- Implementing secure access control and data privacy boundaries, and building fault-tolerant features with a focus on data integrity.
- Evaluating third-party integrations for feasibility, security, and cost, and backing technical decisions with test evidence alongside cross-functional teams.

TheCloudOps — December 2024 – September 2026

Senior MERN Stack Developer / Full-Stack Developer / Team Lead — *Lahore, Punjab*

- Led the end-to-end development of a **Private Jet Aviation Charter Platform**, a full-stack aviation management platform supporting real-time flight booking and management.
- Drove full-stack development of an **Ad-Supported Music Platform** with an 11M+ document catalog, owning AI-powered ad targeting, header-bidding integration, SEO optimization, and large-scale MongoDB migrations.
- Implemented AI-powered features, including AI-driven advertising and intelligent application capabilities, using RAG, embeddings, vector search, tool/function calling, and AI agents.
- Led and mentored the development team through code reviews, technical guidance, and breaking complex requirements into actionable tasks, keeping engineering practices scalable and maintainable.
- Acted as the primary technical point of contact for clients, gathering requirements, translating business needs into technical solutions, and ensuring on-time delivery.

OIOI Group (MARUI) — November 2023 – November 2024

MERN Stack Developer — *Lahore, Punjab*

- Led the complete development of the OIOI Bike Inventory System, streamlining warehouse operations, inventory tracking, stock movement, and transfer management with real-time transfer logging.

- Built secure APIs over a MySQL inventory database, with validation and access controls for creating, updating, retrieving, and deleting inventory records.
- Designed a Node.js microservices architecture connected through REST APIs and RabbitMQ for event-driven communication, packaged in Docker containers behind API gateways with service discovery.
- Deployed and managed the system on AWS (EC2, S3, Lambda) for high availability and scalability, and added Redis caching and real-time data processing to reduce latency.
- Collaborated with cross-functional teams to understand business requirements, troubleshoot technical issues, and deliver maintainable, production-ready software.

KOT Enterprises **September 2022 – October 2023**

Junior MERN Stack Developer *Lahore, Punjab*

- Developed and maintained backend services and RESTful APIs using Node.js and Express.js, supporting scalable and maintainable web application architectures.
- Designed and implemented MongoDB database schemas, models, and queries, ensuring efficient data storage, retrieval, and management across application modules.
- Built and integrated RESTful APIs for core application features, implementing request validation, error handling, authentication, and reliable communication between frontend and backend systems.
- Contributed to the development of responsive and interactive frontend applications using React.js, JavaScript, and Tailwind CSS.

Featured Projects

AI Executive Assistant | *React 19, TypeScript, Node.js, Express.js, Prisma, PostgreSQL, Gemini*

- Building a full-stack AI executive assistant for a company's leadership team, currently in pilot.
- Built document-grounded answers over PDFs, sheets, and images, citing the source file and page.
- Delivered bilingual English and Dutch voice with transcription, check-before-send, and read-aloud.
- Engineered offline-safe reliability: idempotent requests, an offline outbox, and conflict detection.
- Enforced access control on every read and write, scoped to owner, work area, and case, with audit trails.
- Designed a least-privilege Lark tasks and calendar integration, with writes only after explicit approval.

Enterprise Knowledge Intelligence Platform | *React, TypeScript, Python, FastAPI, PostgreSQL, pgvector, Gemini*

- Built an AI knowledge assistant for a private jet charter platform, with answers scoped to each role.
- Implemented an end-to-end RAG pipeline: chunking, embeddings, semantic retrieval, and cited answers.
- Tuned retrieval with relevance thresholds, heading-aware chunking, query rewriting, and retrieval memory.
- Added intent detection, structured outputs, SSE streaming, and tool calling behind authorization gates.
- Built a bounded multi-step agent with step limits, no-progress detection, and step-level tracing.
- Enforced guardrails and cost controls: token budgets, usage monitoring, and backend-only API credentials.

Private Jet Aviation Platform | *React, TypeScript, Node.js, Express.js, MySQL, Stripe, AWS*

- Developed a full-stack private jet charter marketplace connecting operators, brokers, and flyers.
- Engineered Stripe payments end-to-end: authorize-then-capture checkout with SHA-256 idempotency keys, Connect Express operator payouts with instant-payout auto-fallback, and tokenized broker invoicing.
- Designed a concurrency-safe booking engine with 15-minute soft locks preventing double-booking, a 6-step checkout (in-app PDF contract signing, card/wire/balance payments), and owner-approval payment holds.

Ad-Supported Music Platform | *Next.js, TypeScript, NestJS, MongoDB, Redis, BullMQ*

- Developed an ad-supported music lyrics platform serving an 11M+ document catalog (songs, artists, albums) with SEO-optimized ISR/RSC pages, JSON-LD structured data, and Core Web Vitals tuning.
- Engineered AI content pipelines (song meanings, news) using Gemini, Vertex AI batch jobs, and BullMQ; implemented Google Ad Manager key-value targeting and rolled out PubGuru header bidding to production in phases.
- Built an admin CMS with role-based access control, AI article generation and scheduling, a batch-job operations console, and a UGC moderation suite.
- Cut API query times by 83% (300ms to 50ms) by consolidating a 7M+ document catalog from scattered staging collections into 3 core MongoDB collections with indexed ID-based lookups.

Education

Government College University (GCU), Lahore **September 2016 – July 2020**

B.Sc. (Hons.) Physics *Lahore, Pakistan*

- Authored a B.Sc. (Hons.) thesis, *"Different Attempts To Make Quantum Computers and Computation a Reality,"* analyzing how computers process and represent information, from classical architecture (binary, logic gates, transistors, CPU, memory, data encoding) to emerging quantum models (qubits, superposition, entanglement, quantum logic gates).